

CHAPTER 12

Toward a general theory of metacommunity ecology

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12.1 Introduction

The study of community ecology in a spatial context has a rich history with different roots coming from various disciplines, in particular population biology and geography (MacArthur 1972). Community ecology, as a sub-field of ecology, focuses on the diversity and coexistence of species. One of the hallmarks of spatial community ecology is the theory of island biogeography (TIB), elaborated by MacArthur and Wilson (MacArthur and Wilson 1963), based on a first model they published simultaneously with a very similar study by Levins and Heatwole (1963). The aim of the TIB is to explain various observations emerging from the analysis of biodiversity on islands (e.g., area-diversity and distance-to-mainland-diversity relationships). The thesis defended by the TIB (MacArthur and Wilson 1967) is that most of these observations can reasonably be explained from the balance between island colonization and local extinction processes, and these processes in turn are expected to depend on geographic characteristics such as island size or distance to the nearest continental mass. By recognizing colonization and extinction processes as key drivers of biodiversity at biogeographic scales, MacArthur and Wilson i) were among the first to

propose a mechanistic theory to solve the issue of spatial scale of biodiversity and ii) based their theory on stochastic processes, thus implicitly acknowledging that an average diversity of S does not mean that the S same species should inhabit the site considered at all times.

Since the seminal book of MacArthur and Wilson, the theory on spatial community ecology has followed two main routes: i) a “complex system” approach focusing on species and diversity distributions at different spatial scales, following approaches from landscape ecology, in order to grasp the connection from landscape structure to community and ecosystem processes, e.g., Polis et al. (1997); ii) a “parsimonious assumption” approach aiming at explaining general coexistence mechanisms for community ecology without much attention to the details of the landscape, which in its most recent form is now better known as metacommunity ecology (Holyoak et al. 2005). In recent years, the field of spatial ecology has moved towards a more integrative view, merging both approaches together with a stronger focus on spatially structured food webs to explain coexistence mechanisms and the consequences of species diversity on ecosystem functioning and ecological interaction networks (Massol et al. 2011).

The metaecosystem concept and its associated framework (Loreau et al. 2003; Gounand et al. 2018) has notably introduced the ideas that ecological stoichiometry coupled with spatial dynamics might play a very important role in the control of ecosystem productivity (Loreau, 2004; Massol et al. 2011) and limiting factors (resources and natural enemies) shaping species coexistence have to be understood more generally through the prism of biomass fluxes within and among ecosystems (Gravel et al. 2010b,a,b; Massol et al. 2017).

In parallel to the construction of the metapopulation/metacommunity/meta-ecosystem framework, P. Chesson and his collaborators have made crucial contributions to the theory of species coexistence through a completely different approach (Chesson 1994, 2000b,a,b; Snyder and Chesson 2003). In particular, these works have proposed a useful categorization of coexistence mechanisms based on two main classes: on the one hand, some mechanisms can be branded as equalizing because their ultimate effect is to minimize fitness differences between species at large spatial and long temporal scales ; on the other hand, some mechanisms are stabilizing because they tend to increase negative density-dependent interactions within species compared to between-species interactions (also interpreted as niche differentiation). Following the logic of the Lotka–Volterra model, such effects are expected to favor the coexistence of species because they increase the likelihood that the multi-species equilibrium is more stable than the ones lacking one or more species. Some models have implicitly incorporated these ideas into the metacommunity framework through ideas such as the regional similarity hypothesis (Mouquet and Loreau 2002) and the neutral theory of biodiversity (Bell 2000; Hubbel 2001), which basically recasts coexistence theory in the absence of stabilizing mechanisms but with perfect equality between species (Gravel et al. 2006; Adler et al. 2007). Although both equalizing and stabilizing mechanisms work towards species coexistence (i.e., the existence of a feasible equilibrium comprising all considered species), only stabilizing mechanisms actually make this coexistent state resilient to perturbations (Barot and Gignoux 2004). For this reason, the categorization proposed

by (Chesson 2000b) is of particular relevance because it allows for a more precise understanding of the conditions favoring species diversity within a patch and species turnover in space and time.

In this chapter, our first goal is to give the reader a general idea of what metacommunity theory has achieved, what it can explain, and how, and what it has not tackled yet, focusing on theories of community assembly and species coexistence and comparing their predictions. However, going into the details of all predictions and theories linked to metacommunity ecology is way beyond the scope of a single book chapter, see Leibold and Chase (2018) for such an overview). For the purpose of conciseness and consistency, our approach here will be to propose and formalize a single unifying model in the context of patch dynamics (i.e., focusing on species occupancies rather than species abundances) to make predictions on three main statistics (species-species and species-environment correlations as well as spatial species occurrence autocorrelation) that could well capture the distribution shaped by coexistence mechanisms. We will first introduce some theoretical results linked to the Levins metapopulation model and its many offshoots in order to highlight the basic expectations of this model in terms of three correlations describing metacommunity structure: environment-occupancy, first order spatial autocorrelation and co-distribution. We will then resort to simulations to check whether the three proposed statistics are indeed able to discriminate between different scenarios of species coexistence. These results might hopefully guide future empirical and theoretical studies by highlighting relevant signature characteristics of metacommunities as well as weaknesses of occupancy-based metrics, thus kindling future research on appropriate metrics of species coexistence mechanisms.

12.2 A general model for “meta” ecology

We start with the presentation of a single-species model and review elements of metapopulation theory with a unique formalism. The model is developed following the tradition of Levins’ model

of patch dynamics (Levins 1969). Our objective is to provide a tool that could be used to develop the theory further, but that is also coherent with previous tools.

12.2.1 The heritage of the Levins' model of colonization and extinction dynamics

In population and community ecology textbooks, one of the classic models that is often presented to study spatial dynamics and scaling in biodiversity studies is Levins' metapopulation model. Initially proposed in the context of agronomical research, see Levins (1969), this model has been especially fruitful with many extensions, see the reviews in Hanski (1999) and Britton (2013). Interestingly, the model was developed in parallel to MacArthur and Wilson's theory of island biogeography, which also builds on the same fundamental state variable and processes. Essentially, Levins' model describes the dynamics of expected patch occupancy p , which could be interpreted as a fraction of the landscape that is occupied by the species, as well as the probability to find a population at a given location. The metapopulation is dynamic because of the processes of colonization, which is the result of a propagule landing at a location and establishing a viable population, and extinction, which is the result of a permanent disappearance of all individuals from that locality, either because of stochastic population dynamics or a disturbance. An implicit assumption is that the per capita growth rate is positive in all patches, i.e., that the entire landscape is suitable to establish viable local populations (but see following variants).

We start by deriving Levins' model from its underlying processes (colonization and extinction) and then we add spatial heterogeneity. Let us first consider a large landscape of n patches, each of them with approximately the same quality and size, so that colonization and extinction rates are comparable throughout. We look at the probabilities P_k that k of these patches are occupied. Assuming that propagules landing into a patch can come from any other patch (i.e., global dispersal and perfect mixing), this family of probabilities obeys the following master equation:

$$\frac{dP_k}{dt} = \frac{c}{n} (k-1)(n-k+1)P_{k-1} + e(k+1)P_{k+1} - \left[\frac{c}{n} k(n-k) + ek \right] P_k \quad (12.1)$$

i.e., the number of occupied patches can increase if a colonization happens in one of the $n-k$ empty patches (the associated rate being colonization pressure c/n per occupied patch owing to the division of emigrating propagules between the n possible destinations, multiplied by the number of occupied patches, k), or can decrease if an extinction happens (with rate e) in one of the k occupied patches. Multiplying both sides of Equation 12.1 by k/n and summing over k yields the following equation for the dynamics of $p = E[k/n]$, with terms in $v = \text{Var}[k/n]$:

$$\frac{dp}{dt} = cp(1-p) - cv - ep \quad (12.2)$$

If k were a random variable with binomial distribution, we would have $v = p(1-p)/n$, i.e., vanishing variance with increasing n . In general, we can expect v to decrease with n , so, assuming a large number of patches, the dynamics of p can be approximated by Levins' equation:

$$\frac{dp}{dt} = cp(1-p) - ep \quad (12.3)$$

In this form, it is a one-species model with no underlying spatial heterogeneity. The classic results from this model are easy to obtain:

- there are two equilibria for p , either $p=0$ or $p=1-e/c$;
- the first equilibrium is unstable when $c > e$ and the second is stable, which in ecological terms means that persistence occurs if colonization is larger than extinction; the reverse is true when $c < e$;
- the dynamics of this model are strictly identical to those of a logistic growth model with parameters $r = c - e$ and $K = 1 - e/c$, notably a rate of return to the equilibrium equal to $c - e$.

A very natural improvement to the original model stems from the idea that less than 100 per cent of the available patches are fit for species colonization, i.e., the local intrinsic population growth rate is negative in some locations such that establishment is not

feasible even if there are plenty of propagules. The following formalization, introduced by (Nee and May 1992) in a two-species context, adds another parameter to Levins' model, i.e., the proportion of inhabitable patches h :

$$\frac{dp}{dt} = cp(h - p) - ep \quad (12.4)$$

This model does not change results radically for the one-species, one-habitat type model, i.e., there are still two equilibriums (the second becomes $p = h - e/c$), the critical inequality that must be satisfied for the metapopulation to persist is now $hc > e$, and the rate of return to the non-empty equilibrium is $ch - e$. However, the introduction of this new parameter allows one to model habitat heterogeneity and ecological interactions in a simple way.

12.2.2 Local demography versus regional processes

Levins' metapopulation model and its various offshoots are "macroscopic" models insofar as they concentrate on colonization/extinction dynamics and do not treat population dynamics within a patch. An important implicit assumption of this formalism is that local dynamics are happening much faster than regional dynamics, such that they could be ignored from the model (see Gravel et al. (2010a) for relaxation of this assumption). Essentially, a just-colonized patch is assumed to be identical (on average) to a patch that has been occupied for a long time. Population dynamics are assumed to be fast enough, when compared to colonization/extinction dynamics, for newly formed populations to reach their carrying capacity before the next extinction event in the metapopulation. This assumption is obviously not realistic in all metapopulation settings, especially when there is a strong seasonal forcing on population and colonization/extinction dynamics. A good reading on the limits of this assumption is provided by Keeling (2002) and Eriksson (2013).

12.3 Spatial heterogeneity

Coexistence is impossible in the Levins metapopulation model in the absence of spatial heterogeneity or the possibility of local species replacement.

Gause's competitive exclusion principle can easily be reformulated to apply to spatial dynamics by stating that two similar species competing for space cannot coexist. Akin to local dynamics, some form of heterogeneity is required to obtain spatial coexistence. We will see in the next section how differences among species may contribute to it, but first we need to establish how spatial heterogeneity affects single-species metapopulations. This step is necessary to reveal how environmental variation and limited dispersal contribute to affect the equilibrium occupancy of a resident species. In this section we analyze extensions of the single-species Levins model in order to reveal the effects of environmental variation and dispersal limitation on the equilibrium occupancy. Our analysis formalizes how environmental occupancy and spatial correlations may arise and structure species distribution.

12.3.1 Environmental variation

Introducing environmental variation into Levins metapopulation model can be achieved in all generality by making model parameters (i.e., c and e) dependent on patch type and stating the proportion of total patches assigned to each type (i.e., the equivalent of parameter h). For simplicity, consider a metapopulation which comprises two habitat types, say favorable (in proportion π) and unfavorable (in proportion $1 - \pi$). We follow a single species which inhabits this metapopulation, and its metapopulation dynamics are governed by the usual processes of colonization and extinction.

Extinction rate varies between the two habitat types and is equal to e_F in favorable patches and to e_U in unfavorable patches. In the same vein, contribution of occupied favorable patches to colonization (resp. unfavorable) is c_F (resp. c_U). To keep in line with the spirit of approaches proposed earlier (Slatkin 1974; Christiansen and Fenchel 1977; Hanski 1983), unfavorable patches are also considered as potentially more (or less) resistant to colonization through parameter r_U , which describes the probability that colonization can take place.

The differential equations that define the dynamics of species occupancy in both habitat types, (p_U in favorable habitats, p_F in unfavorable ones, and total occupancy $p = p_U + p_F$) according to Levins metapopulation model are:

$$\begin{aligned}\frac{dp_F}{dt} &= (c_F p_F + c_U p_U) (1 - \pi - p_F) - e_F p_F \\ \frac{dp_U}{dt} &= r_U (c_F p_F + c_U p_U) (\pi - p_U) - e_U p_U\end{aligned}\quad (12.5)$$

The main interest of this variant of Levins metapopulation model is that it can predict the expected correlations between species occupancy and patch type, a quantity that we will be interested in later when we will explore statistical moments of metacommunities. To obtain this correlation, we first need to derive expressions for equilibrium occupancies. These are given by the following equations:

$$\begin{aligned}(1 - \pi - p_F) (c_F p_F + c_U p_U) &= e_F p_F \\ r_U (\pi - p_U) (c_F p_F + c_U p_U) &= e_U p_U\end{aligned}\quad (12.6)$$

which can be solved by introducing the variable m defined as:

$$m = 1 / (c_F p_F + c_U p_U) \quad (12.7)$$

yielding p_F and p_U :

$$\begin{aligned}p_F &= \frac{1 - \pi}{1 + m e_F} \\ p_U &= \frac{\pi}{1 + m e_U / r_U}\end{aligned}\quad (12.8)$$

The equation on m obtained through system (12.6) is a quadratic equation with solutions:

$$m_{\pm} = \frac{f_F + f_U - \bar{c} \pm \sqrt{(\bar{c} - f_F - f_U)^2 + 4[c_F(1 - \pi) + c_U \pi f_F - f_F f_U]}}{2[c_F(1 - \pi) f_U + c_U \pi f_F - f_F f_U]} \quad (12.9)$$

where $\bar{c} = c_F(1 - \pi) + c_U \pi$, $f_F = e_F$ and $f_U = e_U / r_U$. p can be recovered from Equation (12.9) by summing the two parts of system (12.8), i.e., as:

$$p = \frac{1 - \pi}{1 + m f_F} + \frac{\pi}{1 + m f_U} \quad (12.10)$$

Plugging Equation (12.9) in Equation (12.10) gives a long formula better left hidden. Figure 12.2 presents a more understandable picture of how p varies as a function of statistical moments of the “effective extinction rates”, f_x , namely its expectation $\bar{f} = \pi f_U + (1 - \pi) f_F$ and its variance $Var(f) = \pi(1 - \pi)(f_U - f_F)^2$. A larger difference between the two effective extinction rates at a given average effective extinction rate (i.e., going up on Figure 12.1) increases occupancy; conversely, at a given difference of effective extinction rate, increasing the average effective extinction rate (i.e., going right on Figure 12.1) decreases occupancy.

Approximating Equation 12.10 at first order in $Cov[f, c]$ and $Var[f]$, we obtain the following Taylor series approximation for p :

$$p \approx 1 - \frac{\bar{f}}{\bar{c}} + \frac{\bar{f} Cov[f, c]}{\bar{c}^3} + \frac{Var[f]}{\bar{c}^2} \quad (12.11)$$

This expression can also be recovered with a large number of patch types defined by their parameters c_x , e_x and r_x , with $f_x = e_x / r_x$, and the proportion of patches of type x , π_x . The equivalent of system (12.5) then becomes:

$$\frac{dp_x}{dt} = r_x (\pi_x - p_x) \sum_y c_y p_y - e_x p_x \quad (12.12)$$

At equilibrium, Equation (12.12) always implies:

$$p_x = \frac{\pi_x}{1 + f_x / \sum_y c_y p_y} \quad (12.13)$$

and, from Equation (12.12) taken for all x ,

$$p = 1 - \frac{\sum_y f_y p_y}{\sum_y c_y p_y} \quad (12.14)$$

Developing the sum at the denominator in terms of $\alpha_x = p_x / \pi_x$ and using Equation (12.13), one obtains:

$$\begin{aligned}\bar{\alpha c} = E \left[\frac{c_x}{1 + \frac{f_x}{\bar{\alpha c}}} \right] &\approx \frac{\bar{c}}{1 + \frac{\bar{f}}{\bar{\alpha c}}} - \frac{Cov[f, c]}{\bar{\alpha c} \left(1 + \frac{\bar{f}}{\bar{\alpha c}}\right)^2} \\ &+ \frac{\bar{c} Var[f]}{\bar{\alpha c}^2 \left(1 + \frac{\bar{f}}{\bar{\alpha c}}\right)^3}\end{aligned}\quad (12.15)$$

Assuming that $\bar{\alpha c}$ can be decomposed as an average term plus terms in $Cov[f, c]$, $Var[c]$ and $Var[f]$, one finds:

$$\bar{\alpha c} \approx \bar{c} - \bar{f} - \frac{Cov[f, c]}{\bar{c}} + \frac{Var[f]}{\bar{c}} \quad (12.16)$$

With the same type of calculations, we can also obtain the following approximation:

$$\bar{\alpha f} \approx \frac{\bar{f}(\bar{c} - \bar{f})}{\bar{c}} + \frac{(2\bar{f} - \bar{c}) Var[f]}{\bar{c}^3} - \frac{\bar{f}^2 Cov[f, c]}{\bar{c}^2} \quad (12.17)$$

Plugging Equations (12.16) and (12.17) into Equation (12.14) and keeping only sensible terms, we finally recover Equation (12.11).

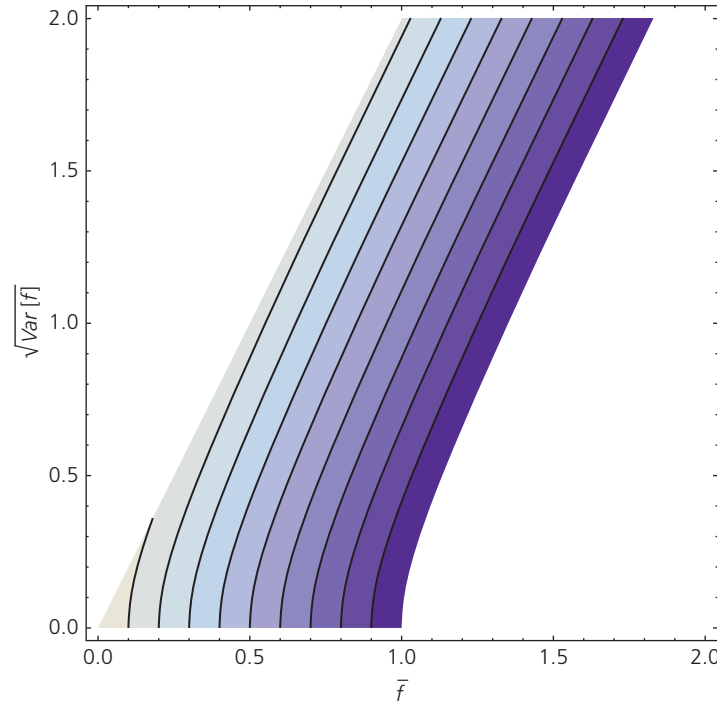


Figure 12.1 Effect of spatial heterogeneity on equilibrium occupancy. The figure illustrates iso-occupancy lines (from 0 to 1, with one line at each 0.1 occupancy increment) as a function of the average effective extinction rate, $\bar{f}$ and the standard deviation of extinction rates, at $\pi = 0.2$, $c_f = c_U = 1$. The area in the top left corner is not possible with only two patch types – it would require negative values for e_f . The area in the bottom right corner leads to $p = 0$.

Based on Equations (12.13) and (12.16), we can also obtain an approximation of p_x :

$$p_x \approx \pi_x \left[\frac{\bar{c} - \bar{f}}{\bar{c} + \Delta f_x} - \frac{(f_x \text{Cov}[f, c])}{\bar{c}(\bar{c} + \Delta f_x)^2} + \frac{f_x \text{Var}[f]}{\bar{c}(\bar{c} + \Delta f_x)^2} \right] \quad (12.18)$$

with $\Delta f_x = f_x - \bar{f}$. Using Equation (12.18), we can get an approximation of the covariance K_x between species occurrence and patch type x occurrence:

$$K_x = p_x - p\pi_x \approx -\frac{\pi_x}{(\bar{c} + \Delta f_x)^2} \left\{ \begin{aligned} &\left(1 - \frac{\bar{f}}{\bar{c}}\right) \text{Var}[f] \\ &+ \left[\left(1 - \frac{\bar{f}}{\bar{c}}\right) \bar{c} + \left(1 - \frac{2\bar{f}}{\bar{c}}\right) \frac{\text{Cov}[f, c]}{\bar{c}} + \frac{\text{Var}[f]}{\bar{c}}\right] \Delta f_x \\ &+ \left[\left(1 - \frac{\bar{f}}{\bar{c}}\right) - \frac{\bar{f}}{\bar{c}} \text{Cov}[f, c] + \text{Var}[f]\right] \frac{(\Delta f_x)^2}{\bar{c}^2} \end{aligned} \right\} \quad (12.19)$$

The correlation between species occurrence and patch type x occurrence, κ_x , can be obtained by dividing the right-hand side of Equation (12.19) by $\sqrt{p(1-p)\pi_x(1-\pi_x)}$ and cutting approximations at the level of variance and covariance terms:

$$\kappa_x \approx -\frac{\sqrt{\pi_x}}{(\bar{c} + \Delta f_x)^2 \sqrt{(1-\pi) \frac{\bar{f}}{\bar{c}} \left(1 - \frac{\bar{f}}{\bar{c}}\right)}} \left\{ \begin{aligned} &\left(1 - \frac{\bar{f}}{\bar{c}}\right) \text{Var}[f] \\ &+ \left[\left(1 - \frac{\bar{f}}{\bar{c}}\right) \bar{c} + \frac{\left(1 - \frac{2\bar{f}}{\bar{c}}\right) \text{Cov}[f, c]}{2\bar{c}} + \frac{\text{Var}[f]}{2\bar{f}}\right] \Delta f_x \\ &- \left[-\left(1 - \frac{\bar{f}}{\bar{c}}\right) + \frac{\text{Cov}[f, c]}{2\bar{c}^2} - \frac{\text{Var}[f]}{2\bar{c}\bar{f}}\right] (\Delta f_x)^2 \end{aligned} \right\} \quad (12.20)$$

Independently of Δf_x , environment-occurrence correlations are expected to be negatively affected

by the variance in effective extinction rates. A patch type with local extinction rate equal to the metapopulation average would be less occupied than predicted from the product of p and π just because of the variance in extinction rates, which creates a deficit in recolonisation of extinct patches. In other words, for a given average performance, spatial variation in the environment reduces the regional performance of a species. Patch types with larger than average f are expected to have more negative habitat-occurrence correlations due to i) the effect of colonization (the higher the colonization rate, the stronger the effect of Δf_x), ii) the variance in effective extinction rates (the higher the variance, the stronger the effect), and (iii) the covariance between colonization and extinction rates depending on whether the extinction- to-colonization ratio is lower or higher than $1/2$. If the ratio is lower than $1/2$, a positive covariance means a negative effect of Δf_x on κ_x ; conversely, if the f/c ratio is above $1/2$, a positive covariance means a negative effect of Δf_x on K_x . The effect of $(\Delta f_x)^2$ is mainly to decrease the value of the correlation (i.e., becoming more negative), with a modulation by $\text{Cov}[f, c]$ if this quantity is positive.

An important result of Equation (12.20) is that when f_x does not vary among patch types, the environment-occurrence correlations all become equal to zero. This is not so surprising: when only colonization rates out of patches differ, we do not expect more or less productive patches to be occupied more or less often if space is modeled implicitly (i.e., we do not take the clustering of patch types into account). Conversely, when only f_x varies but not c_x , the effects of Δf_x and $\text{Var}[f]$ are quite straightforward: the environment-occurrence correlation decreases with the variation in extinction rates, Δf_x , with the product of both terms, $(\Delta f_x)^2$, and also decreases with the product of this term with $\text{Var}[f]$. Since $(\Delta f_x)^2$ and $\text{Var}[f]$ are always non-negative, the only term that can produce a positive correlation is the simple Δf_x effects—patch types with less than average extinction rates should have higher correlations. Because of all the negative effects of $(\Delta f_x)^2$ and $\text{Var}[f]$, this means that positive environment-occurrence correlations are expected to be less intense than the negative ones—the association, positive or negative, is stronger with

unfavorable environment types than with favorable ones. Or, in other words, we expect a negative correlation between species occurrence, and patch x occurrence even when Δf_x is negative, down to negative values of order $-\text{Var}[f]/\bar{c}$.

12.3.2 Dispersal limitation

In this section, we introduce the notion that spatially explicit structure driven by dispersal limitation can induce spatial autocorrelation in species occurrence. Computing the expectation for occurrence autocorrelation is a difficult problem and recent approaches have circumvented this difficulty by applying stochastic differential equation formalism and perturbation analysis (Ovaskainen and Cornell 2006a; Ovaskainen and Cornell 2006b). Here we will focus on a simple toy model that can be approximately solved using pair approximations.

Consider a metapopulation comprising a large number (N) of patches positioned on a circle and assume that colonization processes only take place in a stepping-stone manner (i.e., colonization can only occur from a neighboring patch). We assume that the number of patches is high, so that whole metapopulation extinction due to stochasticity is practically impossible. To analyze the dynamics of such a model metapopulation, we use pair approximations, i.e., we assume that the occurrence of the focal species in a patch can be computed from the knowledge of species occurrence in the two neighboring patches.

To describe the state (occupied, O , or empty, E) of a given patch (with index x) at time t , we use random variable $X_x(t)$. The proportion of patch pairs that are both empty (p_{EE}), both occupied (p_{OO}), or mismatched (p_{EO} and p_{OE}) can be written as follows:

$$\begin{aligned} p_{EE} &= E[(1 - X_x)(1 - X_{x+1})] \\ p_{EO} &= p_{OE} = E[X_x(1 - X_{x+1})] \\ p_{OO} &= E[X_x X_{x+1}] \end{aligned} \quad (12.21)$$

Quite naturally $p_{EE} + p_{OO} + 2p_{OE} = 1$. Species occupancy is then given by $p = p_{OO} + p_{OE}$.

When a patch x is empty, the pair approximation assumption implies that the probability that its left-hand side neighboring patch is occupied is

$p_{OE}/(p_{OE} + p_{EE})$. Based on this assumption and a colonization rate $c/2$ towards each side of an occupied patch, the differential equations governing the dynamics of the metapopulation are given by:

$$\begin{aligned}\frac{p_{EE}}{dt} &= 2ep_{OE} - cp_{EE} \left(\frac{p_{OE}}{p_{OE} + p_{EE}} \right) \\ \frac{p_{OE}}{dt} &= \frac{c}{2}p_{EE} \left(\frac{p_{OE}}{p_{OE} + p_{EE}} \right) + ep_{OO} \\ &\quad - \left[e + \frac{c}{2} \left(1 + \frac{p_{OE}}{p_{OE} + p_{EE}} \right) \right] p_{OE} \\ \frac{p_{OO}}{dt} &= c \left(1 + \frac{p_{OE}}{p_{OE} + p_{EE}} \right) p_{OE} - 2ep_{OO} \quad (12.22)\end{aligned}$$

The equilibrium solution to this set of equations can be explicitly given in terms of e/c :

$$\begin{aligned}p_{EE} &= \frac{2(e/c)^2}{1 - (e/c)} \\ p_{OE} = p_{EO} &= \frac{(e/c) [1 - 2(e/c)]}{1 - (e/c)} \\ p_{OO} &= 1 - 2(e/c) \quad (12.23)\end{aligned}$$

so that metapopulation occupancy $p = p_{OE} + p_{OO}$ is given as:

$$p = \frac{1 - 2(e/c)}{1 - (e/c)} \quad (12.24)$$

and the autocorrelation between neighboring patch states, $\rho = (p_{OO} - p^2) / p(1 - p)$, equals:

$$\rho = \frac{e}{c} \quad (12.25)$$

Plugging $p_{OO} = p^2 + \rho p(1 - p)$ and $p_{OE} = (1 - \rho)p(1 - p)$ into system (12.22), we can deduce:

$$\frac{dp}{dt} = c(1 - \rho)(1 - p)p - ep \quad (12.26)$$

and from Equations (12.26) and system (12.22) :

$$\frac{d\rho}{dt} = c(1 - \rho)^2(1 - p) - e\rho \frac{\rho}{1 - \rho} \quad (12.27)$$

Equations (12.26) and (12.27) yield the equilibrium relationships:

$$\begin{aligned}(1 - \rho)(1 - p) &= e/c \\ (1 - \rho)^2(1 - p)^2 &= \rho e/c \quad (12.28)\end{aligned}$$

from which we can also deduce Equations (12.24) and (12.25).

Equations (12.26) and (12.27) are interesting because they highlight the similarities and differences between the model developed here and the spatially implicit Levins model developed previously. In the spatially explicit version, the effective colonization rate becomes multiplied by $1 - \rho$, which is the mathematical translation of the fact that colonization is most efficient when occupied, and empty patches are intermingled rather than spatially segregated. Spatially explicit dispersal has two consequences that will be helpful to later understand coexistence. First, it reduces the equilibrium occupancy relative to the standard Levins' model with global dispersal model by an amount equal to e/c . Keeping in mind that the extinction rate has to be smaller than the colonization for persistence, it indicates that the negative effect of dispersal limitation will vanish as colonization rate becomes very large, simply because empty patches are rapidly recolonized. Second, the spatial autocorrelation is always positive and also monotonically decline with colonization rate.

12.4 Coexistence and Persistence

12.4.1 Introducing species interactions

We now move to the most general version of a multi-species version of the Levins model. The model is based on the earlier work of several other authors, including Cohen (1970), Levins and Culver (1971), Slatkin (1974), Hanski (1983), Hanski (1999), and Holt (1996). Its parameterization for specific cases, as we will see, and summarized in Table 12.1 covers different situations such as pre-emptive competition (Levins and Culver 1971), competition-colonization trade-off (Tilman 1994; Hastings 1980), predation (Holt 1996; Gravel et al. 2011b) and mutualism (Klausmeier et al. 2001). The following analysis is based on a two-species scenario for tractability, with the two species labelled 1 and 2, but could easily be extended to multiple species.

See Figure 12.2 and Table 12.1 for description of parameters.

Table 12.1 Summary comparison of different parameterization of the two species model for various cases of inter-specific interactions.

		Sp. 1	Sp. 2	Production	Establishment		Extinction		
Ind. metapop.				b_i	$b'_i = b_i$	s_i	$s'_i = s_i$	e_i	$e'_i = e_i$
Competition	Neutral			$b_1 = b_2$	$b'_i = 0$	$s_1 = s_2$	$s'_i = 0$	$e_1 = e_2$	$e'_i = \infty$
	Col. comp.	Sup.	Weak	b_i	$b'_i = b_i$	s_i	$s'_i < s_i$	e_i	$e'_i = e_i$
	Ext. comp.	Sup.	Weak	b_i	$b'_i = b_i$	s_i	$s'_i = s_i$	e_i	$e'_i > e_i$
	CC-Trade off	Sup.	Weak	$b_1 < b_2$	$b'_2 = 0$	$s_1 < s_2$	$s'_2 = 0$	e_i	$e'_2 = \infty$
	Species sort.			Spatially variable					
Pred.-prey	Bottom-up	Prey	Pred.	$b_2 = 0$	$s_2 = 0$	$s_2 = 0$	s'_i	$e_2 = \infty$	e'_i
	Top-down	Prey	Pred.	$b_2 = 0$	$s_2 = 0$	$s_2 = \textit{infty}$	$s'_i > 0$	$e_2 = \infty$	$e'_1 > e_1$
Plant-poll.	Facultative	Plant	Poll.	$b_2 < b'_2$	$b_2 < b'_2$	$s_2 < b'_2$	$s_2 < b'_2$	$e_2 < e'_2$	$e_2 < e'_2$
	Obligatory	Plant	Poll.	$b_2 = 0$	$s_2 = 0$	$s_2 = 0$	s'_i	$e_2 = \textit{infty}$	e'_i

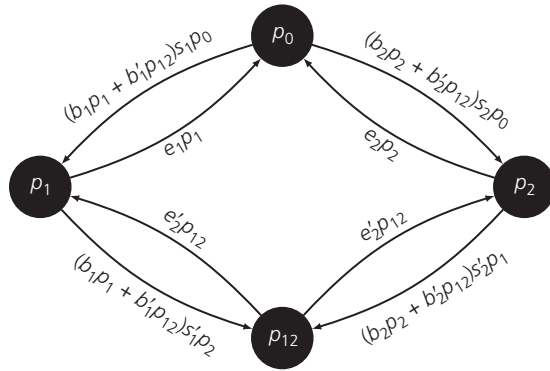


Figure 12.2 Schematic representation of the model.

There are four types of patches to keep track of in a two-species situation: p_1 denotes the fraction of the landscape occupied by species 1 only, p_2 denotes the fraction occupied by species 2 only, p_{12} is the fraction occupied simultaneously by both species and finally, p_0 are empty patches. Following (Hanski 1983), we will simplify notation by defining the quantities $y_1 = p_1 + p_{12}$ and $y_2 = p_2 + p_{12}$. The landscape can only be in one of those states, such that $p_0 = 1 - (p_1 + p_2 + p_{12})$.

Interactions can be represented in various ways, each the result of different local processes. To do so, we first need to decompose colonization in two sub-processes, the propagule production per unit of occupancy, which occurs at rate b_i , and the establishment rate of propagules landing into

a suitable location, which occurs at rate s_i . The traditional definition of an ecological interaction is the effect of a species on the per capita growth rate of another species. If the growth rate is either increased or decreased by the presence of another species, then it should affect the rate at which the population will establish after landing into a patch. This will translate into a modification of the establishment rate in the patches already occupied by the other species, which we will denote s'_i . Since local population dynamics are happening fast relative to regional dynamics, we also need to consider the effect of ecological interactions on the equilibrium population size. Such an effect may either reduce (e.g., in presence of competition or predation) or increase (e.g., in presence of mutualism or a prey) the propagule production, which we will also denote by b'_i . Taken all together, this means that ecological interactions may affect the effective colonization rate in the target patch where establishment occurs and in the source patch from which propagules are being produced. The rates of increase of species 1 occupancy due to the colonization of empty patches and already occupied patches will therefore be $(b_1p_1 + b'_1p_{12})s_1p_0$ and $(b_1p_1 + b'_1p_{12})s'_1p_2$ respectively.

Following the same logic, ecological interactions may also affect extinction rate via both a modification of the equilibrium population size and the capacity of the species to recover from small local perturbations. Again, we denote by a prime symbol

the modified extinction rate e'_i for patches with co-occurrence (p_{12}). The rate of decrease of species 1 occupancy due to its extinction in patches it occupies alone or together with species 2 will therefore be $e_1 p_1$ and $e'_1 p_{12}$ respectively. Note that the rate of joint species extinctions is null.

Collecting the terms together and keeping track of all possible pathways between the different states, we get the following set of differential equations:

$$\begin{aligned}\frac{dp_1}{dt} &= (b_1 p_1 + b'_1 p_{12}) s_1 p_0 - e_1 p_1 + e'_2 p_{12} \\ &\quad - (b_2 p_2 + b'_2 p_{12}) s'_2 p_1 \\ \frac{dp_2}{dt} &= (b_2 p_2 + b'_2 p_{12}) s_2 p_0 - e_2 p_2 + e'_1 p_{12} \\ &\quad - (b_1 p_1 + b'_1 p_{12}) s'_1 p_2 \\ \frac{dp_{12}}{dt} &= (b_1 p_1 + b'_1 p_{12}) s'_1 p_2 + (b_2 p_2 + b'_2 p_{12}) s'_2 p_1 \\ &\quad - (e'_1 + e'_2) p_{12}\end{aligned}\quad (12.29)$$

A schematic version of the model is represented in Figure 12.2 and parameters are interpreted for various settings in Table 12.1. Albeit more difficult to analyze than traditional Lotka–Volterra equations for local dynamics, the model is general enough to account for all combinations of positive and negative interactions. Unfortunately it cannot be solved for internal equilibrium except in special cases, but the mutual invasibility can be studied by stability analysis, following (Hanski 1983), a species can invade a metapopulation equilibrium only comprising the other one if this equilibrium is unstable. This technique has the advantage of highlighting the joint effects of interactions via an analysis of the equilibrium occupancy of the resident species and its effect on the invasibility of the interacting species. Doing so, we will be able to map our interpretation on Chesson's formalism of equalizing and stabilizing mechanisms of coexistence.

12.4.2 Technique for invasibility analysis

It is not possible to solve the two-species model for equilibrium except for a few special cases. It is neither possible to compute directly the per capita growth rate of an invading population because of the p_{12} fraction. The approach to investigate invasibility is therefore to use local stability analysis

on the equilibrium where only one species is resident and the other is completely absent (i.e., $\hat{p}_i = 0$, $\hat{p}_{ij} = 0$ and $\hat{p}_j > 0$). In such a situation, the small perturbation that is the object of the analysis is the introduction of the alternative species and the stability analysis tells us if this small perturbation grows (unstable equilibrium, the species invades) or not (stable equilibrium, resistance to invasion). The following analysis was originally derived by Hanski (1983), but here we introduce a few additions due to the model reformulation and the distinction between propagule production (b_i) and establishment (s_i). We will not detail all derivations here but the reader could refer to Hanski's analysis of regional coexistence.

Stability analysis consists of first computing the Jacobian matrix for the previous set of differential equations, evaluating it at the equilibrium points $\hat{p}_i$ and then investigating the Routh–Hurwitz conditions for stability. After some simplifications of the conditions, we find that local stability depends on two inequalities (setting species 2 at equilibrium, species 1 the invader, and with the following definitions $\mu_i = b'_i s'_i - b_i s_i$):

$$\begin{aligned}b_1 s_1 - b_2 s_2 + (\mu_2 - \mu_1) \hat{p}_2 - 2e_1 - e'_1 - e'_2 &< 0 \\ (b_1 s_1 - e_2 - \mu_1 \hat{p}_2) (b_2 s_2 + e_1 + e'_2 - \mu_2 \hat{p}_2) & \\ < e'_1 [- (b_1 s_1 - e_1) + (b_1 s_1 + b_2 s_2 - \mu_2) \hat{p}_2] &\end{aligned}\quad (12.30)$$

Invasibility will occur if any of these two inequalities is not satisfied. The following analysis for different parameterizations is based on this criterion. Note that coexistence requires this condition to be satisfied at both one-species equilibrium (when each species is resident and the other one absent).

12.4.3 Competition

Fitness inequality and stabilizing interactions

Chesson's theory of coexistence stems directly from MacArthur's heritage formulated in niche theory. Using Lotka–Volterra equations for competitive interactions, it is easy to show that the criteria for stable coexistence depend on the ratio of carrying capacities of the two species and the interaction coefficients among them. Chesson's interpretation of coexistence mechanisms as equalizing and stabilizing directly stems from this condition:

equalizing mechanisms act to reduce the difference in carrying capacities, while stabilizing mechanisms act on the interaction coefficients. For instance, non-linear population dynamics (e.g., a type-II functional response) alter the long-term average carrying capacities via Jensen's inequality, which is interpreted as an equalizing mechanism. Niche differentiation on the other hand reduces the opportunities for species to compete for resources, and thereby reduces the interaction coefficients among them. There is no formal interpretation of these for metacommunities, except by simulations (Roxburgh et al. 2004; Shoemaker and Melbourne 2016).

The previous analysis of equilibrium occupancy with the single-species model was helpful to set-up how different metacommunity settings do affect fitness inequality. The equilibrium occupancy is simply equal to $\hat{p}_i = 1 - e_i/c_i$ for the simple situation where the environment is uniform across the landscape and dispersal is global. Inequality in such a situation therefore only depends on the intrinsic characteristics of the two species. Both environmental variation and dispersal limitations may, however, reduce the equilibrium occupancy, especially when the response to the environment is strong, environmental variation is large, and c_i is low (for a given $\hat{p}_i$). Therefore, species differences in these characteristics may make coexistence more difficult to obtain. (Mouquet and Loreau 2002) already provided a few hints about this situation in their hypothesis of regional similarity, where they conclude that coexistence may be impeded at large dispersal if species are unequal on average across the landscape. Quantification of regional inequality between species is, however, insufficient to conclude about coexistence as we also need to consider the two interspecific interaction coefficients. This is what we will perform in the following analysis of a few cases. In particular, we will look at the boundary conditions between competitive exclusion, unstable coexistence, and stable coexistence over the inequality-interaction space.

Colonization competition (priority effects)

A situation that attracted significant attention of early metacommunity research was the one of

colonization competition, also referred as migration competition by Hanski (1983), where interspecific interactions prevent the establishment of an invading species. It is common to refer at this type of interaction as a priority effect (Fukami 2015), where the resident species establishes and changes the environment so that another species coming to that locality cannot establish any further. In our model, this corresponds to the situation where the establishment rate is lower in the presence of competitor than in empty patches (i.e., $s'_i < s_i$). Note that this situation differs from the one where competition reduces propagule production in patches where species co-occur (i.e., $b'_i < b_i$). In order to compare with Chesson's framework based on niche theory, we consider a gradient of niche overlap σ ranging between 0 (no overlap, perfect differentiation) and 1 (perfect overlap) and that each species establishment rate is reduced with increasing overlap, such that $s'_i = s_i \alpha_i (1 - \sigma)$, where α_i is a positive constant indicating the sensitivity of the species to competition. Note that s'_i has to be non-negative.

One of the main questions asked about colonization competition was if it could prevent coexistence and eventually lead to resistance (alternative stable states). The invasibility conditions, specified by inequality 12.30, simplifies considerably for colonization competition. Invasibility of the resident community *res* by an invading species *inv* is feasible if the following inequality is satisfied:

$$\frac{\hat{p}_{inv}}{\hat{p}_{res}} < \alpha_{inv} \sigma \quad (12.31)$$

As one would guess, it is harder to find stable coexistence with both increasing niche overlap, up to the point where the inferior competitor or the one with the highest occupancy in absence of competition will exclude the other and regain in occupancy (Figure 12.3). Competitive exclusion by the species with the largest regional occupancy will occur if interactions are strong enough. Hanski (1983) also described a situation where priority effects may prevent mutual invasibility and result in unstable coexistence, i.e., a situation where the first species to occupy the landscape resists invasion. This situation is not feasible with the previous definition of competitive interactions, since it is impossible for

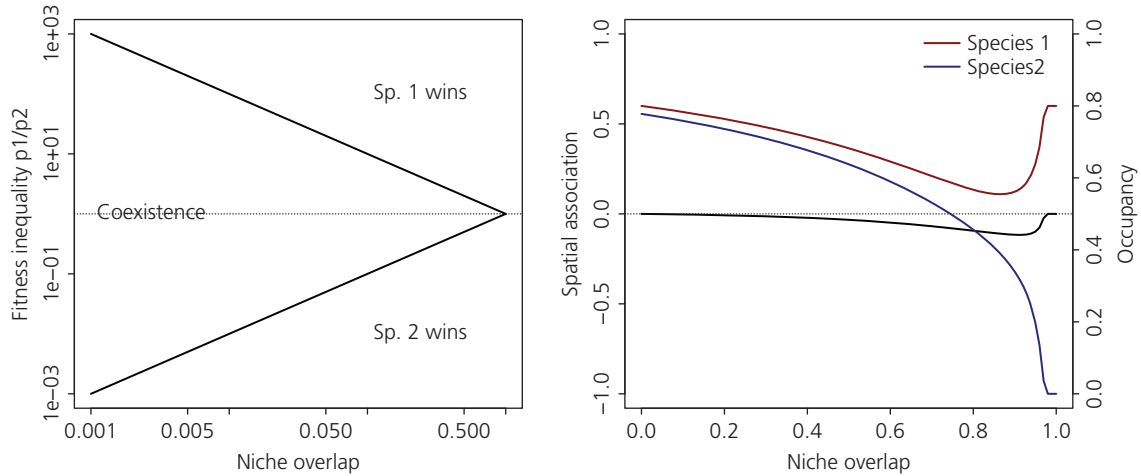


Figure 12.3 Colonization competition. **A)** Regional fitness inequality interaction strength space delimiting where coexistence is feasible. Fitness inequality is computed as the ratio of occupancies of each species in absence of competition. Competition is computed as a function of niche overlap σ such that $s'_i = s_i \alpha_i (1 - \sigma)$, with symmetric sensitivities to niche overlap, $\alpha s_1 = \alpha s_2$. There is no colonization competition at $\sigma = 0$ and it is impossible to colonize occupied patches because of strong priority effect at $\sigma = 1$. Other default parameters are all symmetric except for establishment that is varied to cover the range of occupancies, such that $b_i = b'_i = 1$ and $e_i = e'_i = 1$. **B)** The effect of competition colonization on equilibrium occupancies and on co-occurrence. Here species are slightly unequal by specifying $s_1 = 5$ and $s_2 = 4.5$, other parameters being the same as panel A. Spatial association is the difference between observed co-occurrence and expected co-occurrence under independent distribution, ie $p_{12} - (p_1 + p_{12})(p_2 + p_{12})$.

the two lines to cross on Figure 12.4 (because $\alpha_{inv}\sigma$ could not exceed 1 by definition).

It is interesting to investigate the amount of co-occurrence observed, p_{12} , relative to the expectation if the two species were independently distributed (i.e., $(p_1 + p_{12})(p_2 + p_{12})$), because it may inform on the signature of priority effects on the spatial distribution. Diamond's checkerboard hypothesis (Diamond 1975) arises from the intuition that negatively interacting species will avoid each other in space, which should translate in an observed co-occurrence that is much lower than the random distribution. Diamond's intuition was right, as we see that repulsion increases with interaction strength (Figure 12.4B). That said, the magnitude of the effect is very small relative to the drop in occupancy of both species, in particular of the inferior competitor, such that negative co-occurrence is very unlikely to be detected. One special case of colonization competition that attracted a lot of attention over the years is the neutral model. Hubbell (2001) and Bell (2000) models could be represented in our model by setting all parameters equal and with

strong colonization competition. This situation already attracted the attention of Slatkin (1974) and Hanski (1983), who concluded that coexistence is less and less likely with increasing competition. The zero-sum dynamics represented in Hubbell (2001) model is an extreme case where colonization cannot happen in already occupied patches ($s'_i = 0$). In such a situation the stability of the system is null and therefore make regional abundance subject to a random walk to extinction caused by stochasticity (May 1972; Gravel et al. 2011c). Note that drift is not restricted to this very specific case, it will increase in strength as species inequality reduces and colonization competition increases.

Extinction competition (local exclusion)

Another means by which competitive interactions may affect spatial dynamics is through a modification of the extinction rates. In a situation of pure extinction competition, the two species may colonize all patches, irrespective of their local composition, but then the presence of a competitor may increase the extinction rate, $e'_i = e_i + \alpha_i \sigma$. This

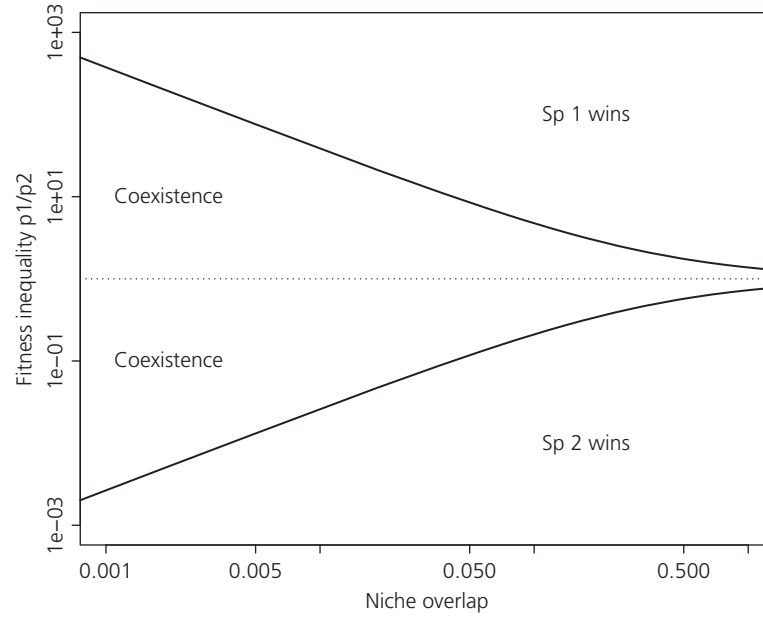


Figure 12.4 Extinction competition. Regional fitness inequality - interaction strength space delimiting where coexistence is feasible under extinction competition. Fitness inequality is computed as the ratio of occupancies of each species in absence of competition. Competition is assumed to be symmetric and correspond to a scaling factor I such that $e'_i = e_i + \alpha_i \sigma$, indicating there is instantaneous extinction of the weaker competitor at $\sigma = 1$. Other default parameters are all symmetric except for establishment that is varied to cover the range of occupancies, such that $b_i = b'_i = 1$ and $e_i = e'_i = 1$.

phenomenon can easily be explained by a reduction in local population size following exploitative or apparent competition, combined with a negative relationship between extinction rate and population size. While the invasion criteria differs significantly from colonization competition, the interpretation of the model is much the same (Figure 12.5). Coexistence is possible if the difference in the occupancies is not too large and competitive interactions not too strong. (Hanski 1983) has shown that priority effects and unstable coexistence are also possible when occupancies are of similar magnitude and interactions very strong, contrary to the original conclusion of (Slatkin 1974).

Competition-colonization trade-off

It is quite challenging to combine both colonization and extinction competition in a single model, but an interesting situation that attracted a lot of attention is one of a trade-off between competitive ability and colonization. While much

credit for this analysis has been awarded to Hastings (1980) and later to Tilman (1994), it is possible to find mentions of it in Skellam (1951) and also in Hanski (1983). Subsequent analyses are found in Pacala and Rees (1998) and Calcagno et al. (2006). The idea is fairly simple and is easily transposed in the model. Basically, one species (labeled species 1) is a strong competitor and therefore could colonize both empty patches and the ones already occupied by the other species, while it is not affected by the presence of the other species. This situation translates in $s_1 = s'_1$ and $e_1 = e'_1$. The other species is a good colonizer of empty patches, but cannot colonize occupied ones, neither defend the ones it already occupies. In such a situation, $s_2 > s_1$ but $s'_2 = 0$, $e_2 < e_1$ and $e'_2 \rightarrow \infty$. Coexistence is feasible with this parameterization if the inequality,

$$\frac{s_2}{s_1} > \frac{\hat{p}_1}{1 - \hat{p}_1} + \frac{e_2}{e_1} \quad (12.32)$$

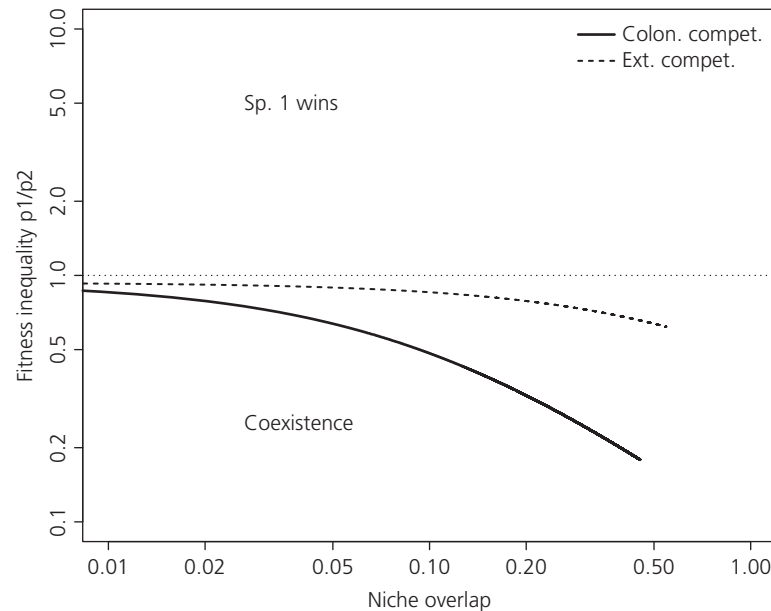


Figure 12.5 Coexistence under the competition-colonization trade-off.

is satisfied. This inequality shows that coexistence is promoted by increasing difference in occupancies between competitors.

What has never been done, however, is its formal interpretation in Chesson's space of fitness inequality and interaction strength, and in particular with less stringent assumption about competitive hierarchies. The previous inequality assumes implicitly that competition is unidirectional (against species 2 only) and infinite (the weaker competitor goes extinct instantaneously). Shoemaker and Melbourne (2016) attempted to partition equalizing and stabilizing mechanisms using a combination of stochastic simulations and time series analysis, but the model they used differs significantly from the Tilman–Hastings framework and therefore a formal decomposition of coexistence mechanisms has never been performed. It is nonetheless possible to perform this analysis using a graphical representation of Hanski's analysis (see Figure 12.6 and inequality 12.30). Interestingly, the model formulation allows us to relax some of the most contested assumptions in the Tilman–Hastings model, namely the perfect pre-emption by the strong competitor and the instantaneous

extinction of the weak competitor. While a satisfying first approximation to a trade-off that is commonly observed in nature, these assumptions prevent a more continuous analysis of the effect of interaction strength on coexistence.

The competition-colonization trade-off is a regional coexistence mechanism based on both fitness inequality and species differentiation. As one would guess from inequality (12.30), traits (e.g., energetic investment in seed production) and spatial conditions (e.g., connectivity of the landscape) leading to a strong inequality in occupancies, in favour of the weaker competitor, will promote coexistence. This interpretation contrasts with the general view that stabilizing processes are necessary to promote coexistence and equalizing mechanisms not sufficient. We find that the stronger the niche overlap, the larger the required amount of inequality between species to allow coexistence of the weaker competitor (species 2). In addition, traditional niche differentiation mechanisms reducing the strength of interspecific interactions also promote coexistence. Relaxing the assumption of perfect pre-emption by the stronger competitor and instantaneous

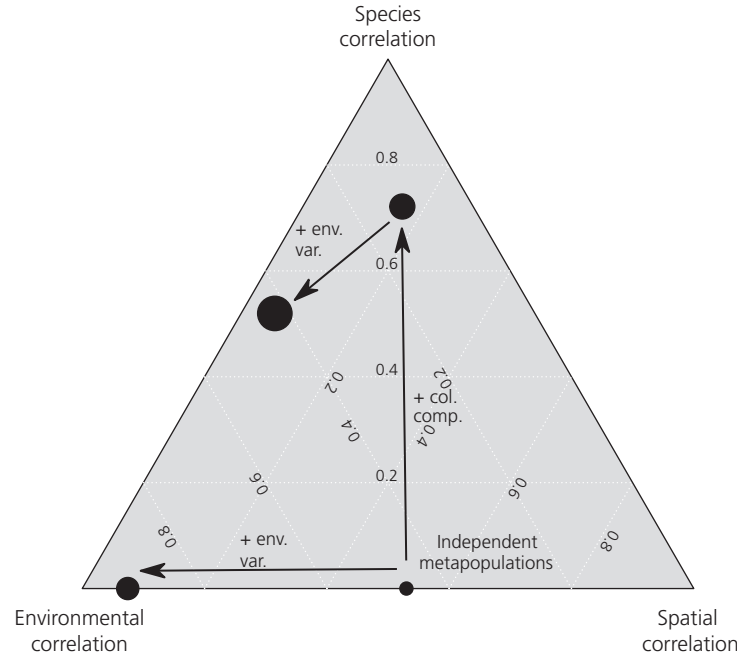


Figure 12.6 Moments for competitive metacommunities. Each dot represents a metacommunity and its size is proportional to the average correlations among the three moments. Species 1 is the strongest regional competitor (because of a higher occupancy). Parameters are: $f_i = f'_i = 1$, $s_1 = 5$, $s_2 = 5$, $s'_i = s_i * (1 - l)$, $e_1 = e'_1 = 1 + d$ and $e_2 = e'_2 = 1 - d$, and $l = 0.85$.

extinction of the weaker competitor both reduce the inequality in occupancies required to promote coexistence. Fitness inequality alone can promote coexistence because there always exists a critical value of s_2 that would allow both species to invade each other, regardless of extinction coefficients (even when $e'_1 = 0$ and $e'_1 = \infty$, in other words when there is immediate extinction of the weak competitor), as long as $\hat{p}_1 < 1$ (Hanski 1983).

Species sorting

Analytical results showing how species sorting may contribute to coexistence with the previous model are impossible to derive, but the few basic principles responsible for the species sorting are easy to understand from the previous findings. Let consider the situation where two competitors have different performances throughout the landscape, either because they respond to different environmental variables or because they differ in their response to a single variable. First, we have previously shown for the single species analysis that spatial variation

in the colonization-extinction ratio for each species will decrease their occupancy. This effect of spatial heterogeneity will be further magnified by dispersal limitation and together they may influence regional inequality. Spatial variation of the environment alone may not have an impact on coexistence if it affects occupancies similarly because the inequality will stay unchanged, but it may promote it if it harms more a specialist competitor (which will experience larger spatial variance) than a generalist but weaker competitor. Differences in niche breadth will move away the species pair from the fitness equality line in the coexistence plane. In addition, spatial heterogeneity will reduce opportunities for interactions. Co-occurrence (p_{12}) will be reduced because of spatial sorting and so will the effective importance of exclusion competition. Further, a negative correlation between patches occupied by a species and the colonization rate of the other species ($cor(p_i, s_j)$) will further reduce opportunities for co-occurrence and therefore competitive exclusion. Overall, conditions for coexistence will

be made easier with a move toward the left and the center in the inequality/interaction space.

12.4.4 Food webs

The persistence of spatial food webs can also be investigated using the same model. Driven by a simple set of principles, established first by Holt (1996) and Holt (2005) and later generalized by (Calcagno et al. 2011), Gravel et al. (2011b), and Massol et al. (2017). The basic model considers a sequential assembly of a linear food chain made of specialized species, with the bottom-level prey establishing first, followed by a predator and eventually by a top predator. The entire food chain may collapse with the extinction of a low-level species. In the most extreme situation, the predator (labeled species 2) cannot colonize a patch where the prey (labeled species 1) is absent and it goes automatically extinct with the extinction of the prey. These assumptions translate into the model with $s_2 = 0$ and $e_2 = \infty$, but they could be relaxed if one considers that the predator may forage and stay for a short amount of time in unoccupied locations. The spatial dynamics are bottom-up driven in the most simple formulations and therefore $s_1 = s'_1$.

The criterion for the persistence of the prey is the same as an independent metapopulation in the situation where the predator does not harm the dynamics of the prey, i.e., $s_1 - e_1 > 0$. The predator will invade if the following inequality is satisfied:

$$c_2 p_1 > e_2 \quad (12.33)$$

which indicates that it will persist only if the prey occupancy is large enough relative to the predator extinction rate. The equilibrium occupancy of specialized predators ($p_2 = p_1 - e_2/c_2$) will always be lower than the one of their prey, which eventually will put a limit on the number of trophic levels that can be observed in a region, unless the colonization-extinction ratio scales with trophic level—Holt (2005). A consequence of these assumptions is that the predator will always co-occur with the prey, but the opposite will not always be true. Interestingly, the invasion criteria will remain the same in presence of top-down regulation, where the presence of the predator increases the extinction of the prey ($e'_1 > e_1$). Occupancies of both the

prey and the predator will, however, be reduced with top-down regulation, and while the predator distribution will remain nested within the one of the prey, the distribution of the prey will be more negatively associated with the one of the predator.

While much of the focus in this section is about two-species mutual invasibility, it is relevant to note a few general results that have been derived for more diverse communities. Persistence of a predator will increase with the generality of its diet, as it is easier to find prey in empty patches and for interactions to rewire following species extinctions (Gravel et al. 2011b). Further, the topology of the regional food web is also affecting the occupancy of the different species because of high order interactions. A predator specializing on a prey with a large occupancy is more likely to persist than a predator selecting a prey with smaller occupancy (Massol et al. 2017). The co-distribution of multiple prey species is also influencing the occupancy of the predator as it affects the total fraction of the landscape that is suitable for colonization (Gravel et al. 2011a). Negative association between prey will increase the persistence of the top predator because of a larger total amount of suitable patches, while positive association will reduce it. On the other hand, in the presence of top-down regulation (a positive effect of the predator on the prey extinction rate), apparent competition may appear at the regional level, reducing total prey occupancy and generating negative association between them (Holt 2005). The effect of spatial variation in the environment on food web assembly has yet to be studied, but one could easily predict that a positive covariance in the response to the environment between the predator and the prey (i.e., species share the same abiotic niche) will promote the persistence of the predator (Cazelles et al. 2016). More discussion about the impact of spatial dynamics on food web structure and stability is provided as a perspective at the end of the chapter.

12.4.5 Mutualism

The assembly of mutualistic communities follows essentially the same principles as bottom-up predator-prey dynamics (Fortuna and Bascompte

2006; Astegiano et al. 2015), with the distinction that interactions may be mutually beneficial (versus unidirectional for predator-prey) and that mutualists are often facultative. A plant, for instance, will have a much larger colonization rate and potentially a smaller extinction rate in the presence of its mutualist. A specialized pollinator may, however, go extinct from a region if its partner is not abundant enough. In the case of a chain of interactions, it may be expected to see amplification of occupancies, rather than an inefficiency such as the one observed for predators.

12.5 Moments of metacommunities

We now turn to numeric simulations to investigate potential interactions among the different assembly mechanisms that were previously discussed. Simulations also facilitate the investigation of the three moments of metacommunities. The simulation model applies to a large number of patches the two-species model described previously, with the addition of spatial variation in the environment and spatially explicit dispersal. We summarize the spatial structure of metacommunities by looking at three measures of correlation investigated previously: i) **Environmental correlation** reports the spatial association between the occurrence of each species and the environment. Here, because we solve the model for equilibrium occupancies, we compute it as the Pearson correlation between the environment at each patch, E_x , and the expected occupancy, $P_{ix} = p_{ix} + p_{ijx}$. ii) **Spatial correlation** reports the first order spatial autocorrelation between patches. It is computed as the Pearson correlation between occupancies of nearest neighbour patches. iii) **Species correlation** reports the amount of spatial association between species relative to an independent distribution. It is computed as the scaled difference between expected co-occupancy and the independent expectation, $(p_{12} - P_1P_2)/P_1P_2$, across all patches. Note that these representations, based on correlations, do not account for shared responses; more sophisticated techniques may solve for partial correlations and allow a complete partitioning of the variation (Ovaskainen et al. 2017; Leibold et al., in review). Here we focus on the correlations to be coherent with previous analytical derivations.

We expect to see situations where synergies among mechanisms could magnify the importance of a given moment, or alternatively antagonism among them may occur. For instance, competitive interactions may force species to occur at locations that are most favourable to them, thereby increasing the environmental correlation. On the other hand, predator-prey interactions may reduce environmental correlation and increase species correlation when the two species share different environmental requirements. We also expect these interactions to be dependent on the spatial contingencies, such as the amount of variance and spatial autocorrelation in the environment and the structure of the connectivity matrix. It was shown for instance that the strength of species sorting in competitive metacommunities may be affected by the landscape structure since short boundaries between environmental types may induce a mass effect as well as dispersal limitations (Ai et al. 2013; Fournier et al. 2017).

We conducted spatially explicit numerical simulations of the model represented in Figure 12.2. We considered a landscape made of $n = 25$ patches. The landscape is simulated as a random geometric graph by distributing patches randomly in a square plane and connections made between the ones distant by less than a threshold value r . For each patch x we consider a local environmental condition E_x that has an average of 0 and a random deviate d_x . For simplicity, and to respect the previous analytical derivations, we consider there are only two types of patches (favorable and unfavorable), and that the local environment only influences equally the extinction parameters e_i and e'_i . We solve the model numerically for equilibrium occupancies for all four types and each patch.

12.5.1 Competition

We simulated four scenarios to illustrate interactions among dispersal limitation, environmental heterogeneity and competitive interactions (here limited to colonization competition, but the same results would hold under extinction competition). The basic simulation starts with two independent metapopulations with distinct connectivity landscapes and no underlying environmental variation

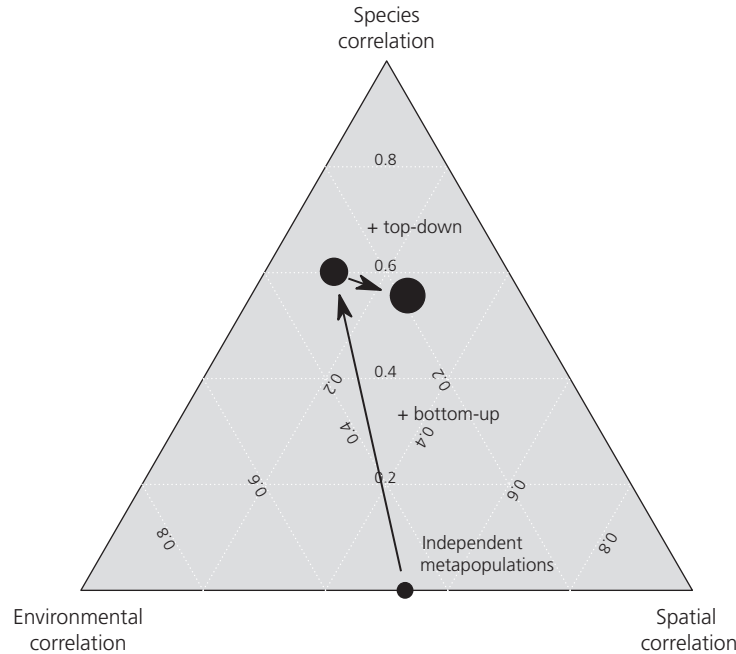


Figure 12.7 Moments for predator-prey interactions. Each dot represents a metacommunity and its size is proportional to the average correlations among the three moments. Species 1 is the predator and species 2 is the prey. The environment is spatially uniform. Parameters are: $f_i = f'_i = 1$, $s_1 = 0$, $s_2 = 4$, $s'_1 = 0$, $s'_2 = 4$, $e_1 = 1000$, $e'_1 = 5$, $e_2 = 1$, $e'_2 = 1$ for the bottom-up simulation and $e'_2 = e_2 = 1/(1 - l)$ for the top-down situation, where $l = 0.7$.

(Figure 12.7). The position of the metacommunity tends to be toward the bottom right of the triangle, dominated by spatial autocorrelation. The exact position of the metacommunity may vary with parameterization and the spatial structure of the landscape, but essentially the only significant correlation is spatial. Some spurious correlation with the environment and between species may, however, happen by chance alone, see Bell (2005), but it will tend to vanish with an increasing number of patches, while it will increase with the amount of spatial autocorrelation in the environment. Not surprisingly, adding variation in the environment moves the location of the metacommunity toward the left of the triangle (species sorting). Alternatively, adding competitive interactions moves the location of the metacommunity toward the top of the triangle, with increasing species correlation. As it would be expected, combining environmental heterogeneity and competition moves the metacommunity toward a more central position; interestingly, the correlations are also

stronger, indicating that both species sorting and negative spatial association get stronger when acting together. Stochastic simulations of a similar model also suggest that some minimal amount of species sorting is required to obtain species correlation because otherwise, competitive exclusion drives spatial distribution and blur any spatial correlation (Leibold and Chase 2018).

12.5.2 Predator-prey interactions

The previous section on coexistence has shown that species correlation may be stronger in the situation of predator-prey interactions because the distribution of specialized predators is nested within the one of their prey. We start again with the same simulation as previously made of two independent metapopulations over a spatially uniform environment. The dependence of the predator on the presence of the prey strongly increases the species correlation. Interestingly, the situation also prevails even if the interaction is

strongly asymmetric with top-down regulation: the predator and the prey maintain a positive spatial association, despite the negative effect of the predator on the prey. The distribution of the predator is, however, completely nested within the one of the prey (i.e., $y_1 = p_{12}$ and $p_1 = 0$), while the prey is much more often alone than with the predator. This distinction is important for co-occurrence analysis and we underline that asymmetric indices are much more appropriate to investigate assembly processes and their ensuing patterns (Cazelles et al. 2015).

12.6 Discussion

Our objective in this chapter was to propose a single model unifying different theories of metacommunity assembly, from competitive interactions to food webs. The model we presented is a re-interpretation of the one analyzed by Hanski (1983) in his investigation of coexistence in competitive metacommunities, following the earlier work of Cohen (1970), Levins and Culver (1971), and Slatkin (1974). Previous metacommunity theory was based on a wide range of modelling approaches, such as spatialized Lotka–Volterra equations, e.g., Takeuchi (1989); Neuhauser and Pacala (1999), multi-species Levins models, e.g., Hastings (1980); Tilman (1994); Mouquet and Loreau (2002); Holt (2005); Gravel et al. (2011a), individual based models, e.g., Bell (2000); Chave et al. (2002), Markov-chain island-mainland models (Hubbell 2001), point-pattern analysis (Bolker and Pacala 1997), and moment approximations (Chesson, 2000a; Snyder and Chesson 2004). Leibold et al. (2004) proposed a unification of different theories for competitive metacommunities with the definition of four paradigms: neutral, species-sorting, mass effect, and competition-colonization trade-off. The proposition was based on a synthesis of the different theories at that time and free of any modelling approach. This contribution was fundamental to the advancement of the field but now we see the limitations of a verbal theory arising. These include the appreciation that the four paradigms are extreme cases of a more continuous reality (Logue et al. 2011); mechanisms are not exclusive, they can be additive or they could interact (Leibold and Chase 2018); contingencies

such as landscape structure, disturbances, or environmental variation may affect which mechanism dominates community assembly (Ai et al. 2013); species not only interact by competition, they are embedded in a more diversified network of ecological interactions (Cazelles 2016); species are not all equal, and their distributions may be influenced by different processes (Fournier et al. 2017); metacommunity theory needs to provide a wider range of predictions than the traditional opposition between spatial autocorrelation and species-environment correlation (Leibold et al., in review). Hopefully, the model we presented here will prove general enough to solve at least part of these problems, and be used as a foundation for a new phase of metacommunity research.

A single mathematical formalism allows us to underline the key parameters to investigate, in theory but also in the field. We placed our analysis of coexistence in a fitness inequality and niche overlap space in order to compare regional versus local coexistence mechanisms. We quantified fitness inequality as the ratio of metapopulation carrying capacities (the equilibrium single species occupancy). Equilibrium occupancy is a function of the ratio of extinction to colonization rates in spatially homogeneous environments and we have shown how it is reduced by both spatial variation of the environment and spatially explicit limited dispersal. This quantity proves to be essential to understand coexistence of competitors in a spatially heterogeneous environment, but also bottom-up assembly of food chains. As an extension of the definition for local dynamics, we also define interactions at the regional level as the per patch impact of a species on the colonization and the extinction rates. Indeed, niche overlap reduces local per capita growth rate, a process scaling up at the regional level by its effect on the colonization rate. Similarly, niche overlap reduces equilibrium population size and eventually promotes extinction.

We also proposed a set of three emerging properties to characterize the structure of metacommunities. More specifically, we looked at three moments of the multivariate species distribution: i) first-order spatial autocorrelation is a measure of the similarity in species composition among adjacent localities and is mostly driven by the

joint effects of disturbances (extinctions) and dispersal limitations; ii) environment-occurrence correlation is a measure of the association between species distribution and the abiotic environment resulting from variation in fitness; iii) species co-distribution is a measure of the spatial association between species and results from various causes, including biotic interactions, shared environmental requirements, and dispersal limitations. Gilbert and Lechowicz (2004) and Cottenie (2005) were the first to describe metacommunity structure with spatial autocorrelation and environment-occurrence correlation. More recently, the development of joint species distribution models (Pollock et al. 2014; Warton 2015) facilitated the analysis of co-distribution and opened new ways to investigate metacommunity dynamics (Zurell et al. 2018). The analysis of co-distribution has a long and rich history in community ecology but was mostly ignored from metacommunity research, but see Bell (2005). Biotic interactions are included in all of the four paradigms of metacommunity dynamics of Leibold et al. (2004), but surprisingly they are not the object of predictions to discriminate them. Various forms of competitive interactions should leave an imprint on species distribution, beyond the checkerboard distribution proposed by Diamond (1975).

Our analysis was not meant to be extensive and future work is, however, needed to identify the circumstances leading to a distinctive co-distribution. We have shown, for instance, that symmetric competition hardly leads to negative spatial association between species, in opposition to Diamond (1975) hypothesis. The onset of competitive exclusion occurs much before negative correlation can be detected from empirical data. We found negative association between competitively interacting species, but measuring this signal in noisy empirical data may require sampling effort much beyond what is typically accessible. Joint species distribution models are powerful tools to quantify co-distribution but they are also limited to the special case of symmetric associations. Asymmetric interactions, in particular antagonistic ones, are impossible to document using covariance or joint probabilities. Rather, conditional probabilities may prove to be more flexible

(and realistic) to describe associations (Cazelles et al. 2015). Shared environmental requirements, indirect interactions, and diffuse interactions are all conditions that are susceptible to mask spatial associations, or alternatively to generate false signal. Investigation of co-distribution in metacommunities should, therefore, be the main object of future metacommunity theory.

Not only does the model we proposed allow development of theory, it could be tested explicitly with measurements of transitions between the different species composition. This analysis was done recently in an investigation of range shifts of temperate forests in North Eastern America (Vissault et al. in review). Forest inventory data was used to document transitions from presence to absence (extinction) and absence to presence (colonization). The model was parameterized across North America and all transitions were conditioned on annual average temperature and precipitations. All of the previous coefficients of species interactions were successfully evaluated, and solving the model made it possible to investigate conditions where competitive interactions limit species range. The model was fitted from dynamical data (transitions) and solved at equilibrium to successfully predict static distribution (current ranges) of both temperate and boreal tree forest. The model was then used to show that stability is the highest in the middle zone, where temperate tree species coexist with boreal ones, and minimal at range limits where interactions are driving one species to the extinction. Trees make a good case for testing such a theory because their spatial dynamics are often driven by disturbances (e.g., canopy gaps, fires) and succession, and because they are also strongly dispersal limited (Hubbell et al. 1999).

12.6.1 Extension to multi-species communities

We focused on pairwise interactions in this chapter but the model we proposed is easy to scale up to entire communities embedded in a diverse network of interactions. The key problem is to represent how interactions add up and influence species dynamics. Tradition in local community ecology is to consider that multiple interactions are additive. In Lotka–Volterra equations, for instance,

interactions are summed over the community and a given species has the same effect whether it is alone or in an entire community. If interactions are additive, then it is possible to approximate the total effect of interactions with knowledge of average interaction strength and the covariance between interaction strength and occupancy (Godsoe et al. 2017). Multiple interactions are also additive in consumer resource models (Brose et al. 2006), although they may influence rate of consumption for Type II and Type III functional response and generate non-linearities. Cazelles et al. (2016) rather introduced the idea that the effect of other species saturates with increasing diversity in a model of spatial dynamics similar to the one considered here, with the first interaction being the most significant and the subsequent weaker. At the end of the spectrum, (Gravel et al. 2011c) proposed that only the first interaction matter for spatial dynamics, allowing the colonization and the persistence of a consumer if at least one prey is present.

Multi-species analyses have shown that a network approach may be the way to reduce the complexity of spatial community dynamics. With the assumption that a predator requires at least one prey to colonize an area and to persist, one can easily derive the expectation for occupancy as a function of the generality of the diet (Gravel et al. 2011c). As a consequence, generalists are expected to be the first to colonize a new location, documented empirically by Piechnik et al. (2008) and to persist. This per species effect scales up and influences how local network properties and body size distribution are affected by area and isolation (Jacquet et al. 2017). Another important aspect to consider in multi-species communities is the effect of higher-order interactions. A species at the top of the food chain not only needs to find its prey, it will be limited by the distribution of the the prey of its prey; as a result, extinction cascades can occur and influence regional food web dynamics (Holt et al. 1999; Calcagno et al. 2011; Massol et al. 2017). Such a theory has not been explored for competitive interactions, but some principles will nonetheless apply. Generality of a species will influence its persistence in competitive metacommunities, although a reversal should be expected, with the least connected

species being more persistent and the first to colonize empty locations. Further, higher order interactions may also be important for occupancy because of the “enemy-of-my-enemy-is-my-friend” principle.

A set of simple rules also explains how co-distribution scales from a few species to entire networks (Cazelles et al. 2015): i) spatial associations may arise from indirect interactions; ii) associations do not have to be symmetrical; iii) multiple interactions decrease the strength of pairwise associations. Putting these rules together, it was predicted that pairwise associations should tend toward randomness with increasing degree and topological distance between species. As a consequence, species are expected to be independently distributed in species-rich communities. In such a situation, we should expect to find stronger environment-occurrence correlation as well as spatial autocorrelation.

12.6.2 From coexistence to dynamical stability

While species coexistence is a central question of community ecology, another closely connected topic is the issue of community stability. Stability has been defined in a variety of ways, reviewed e.g., by Donohue et al. (2013), and theoretically explored by (Arnoldi et al. 2016), but the central tenet of a stable (meta)community remains the same across definitions: rapidly returning to equilibrium after a perturbation. Coexistence and stability are linked in multiple ways, e.g., since coexistence of species at an equilibrium means that this equilibrium is stable (but all coexistence situations are not at equilibrium), and because coexistence can also be recast as mutual invasibility of all communities lacking one species, which amounts to looking for n communities of $n - 1$ species that are unstable to invasions by the n^{th} species.

The question of ecosystem stability is an old-time classic in ecology, dating back at least to (MacArthur 1955). In a provocative paper, (May 1972) challenged the view that species diversity and ecological network complexity contribute positively to ecosystem stability by providing a null model for community stability. He proposed to use random Jacobian matrices to assess whether a randomly

assembled set of S species, interacting with connectance c and with Jacobian elements of mean 0 and variance σ^2 , could represent a stable ecosystem. The answer was that such a system imposes a strong constraint on all of these factors (May 1972; Allesina and Tang 2012), i.e., $\sigma\sqrt{Sc} < m$ where m is the value of diagonal elements of the Jacobian matrix (feedbacks of species upon themselves). (Allesina and Tang 2012) further elaborated on this theory by providing sophistications of the model, taking into account the proportions of predator-prey mutualistic and competitive interactions, and making a distinction between asymptotic stability and reactivity (Tang and Allesina 2014).

In the same vein, Gravel et al. (2016) explored the question of meta-ecosystem stability using the same framework as the one initially envisaged by May (1972), see also (Coyte et al. 2015 and (Mougi and Konodh 2016). By decomposing links between populations as interactions and dispersal rates, Gravel et al. (2016) evinced a major stabilizing factor arising from spatial structure when dispersal is sufficiently high and patches behave heterogeneously. In this context, dispersal mixes heterogeneous random Jacobian elements together, and these random variables are thus averaged over space, with a sample variance decreasing with sample size (i.e., the number of patches averaged over). With a reduced variance of the average ecosystem, stability is enhanced—in May (1972), the σ that appears in the stability criterion is divided by the number of patches that are being averaged. More generally, even when patch-specific Jacobian elements cannot be considered independent random variables, the σ is still divided by the effective number of independent patches in the meta-ecosystem, this quantity being defined through the number of patches and the correlation of their Jacobians. By simulating spatially structured Lotka–Volterra competitive systems, Gravel et al. (2016) were finally able to show that the result obtained through random matrix modelling also held in random dynamical systems, provided dispersal is not too high—at very high dispersal, patches tend to synchronize and thus, the effective number of independent patches decreases with dispersal.

In the case of the multi-species version of Levins' metapopulation model, stability can be assessed

using the same kind of theory. Providing general stability assessment of such models is beyond the scope of this chapter, but we can provide a simple example based on the competitive metacommunity framework used by Calcagno et al. (2006). In this model, each patch can only be occupied by a single species, and occupancy of species i , p_i , is governed by its colonization rate, c_i , the general extinction rate, e , and all the probabilities of successful replacement of species l by species k , η_{kl} :

$$\frac{dp_i}{dt} = c_i \left(h - \sum_j p_j \right) p_i - ep_i + \sum_j (c_i \eta_{ij} - c_j \eta_{ji}) p_i p_j \quad (12.34)$$

If all species are assumed to be present (i.e., non-extinct at the metacommunity scale), the computation of the Jacobian matrix elements J_{ij} is highly simplified:

$$J_{ij} = -c_i p_i^* + (c_i \eta_{ij} - c_j \eta_{ji}) p_i^* \quad (12.35)$$

where p_i^* denotes equilibrium occupancies. Following (Tang and Allesina 2014), asymptotic stability of this system, if its Jacobian elements were random, could be obtained by looking at the leading eigenvalue of the Jacobian matrix, as the maximum of $(S-1)E - d$ and $(1+\rho)\sqrt{SV} - E - d$, where E is the mean of non-diagonal Jacobian elements, $-d$ is the mean of diagonal Jacobian elements, V is the variance of non-diagonal Jacobian elements, and ρ is the correlation between Jacobian elements across the diagonal. Since $E < 0$ and $d > 0$ in this system, the leading eigenvalue of a random Jacobian would thus be $(1+\rho)\sqrt{SV} - E - d$. However, for the system to be stable, this would mean that $-E - d$ should at least be negative (otherwise, there is no way the leading eigenvalue could have negative real parts). Using the symmetry of the $c_i \eta_{ij}$ terms in the Jacobian and the equilibrium conditions given by setting the right-hand side of Equation 20.34 equal to zero with all $p_i > 0$, it can be shown that $-E - d$ tends towards $\langle c_i \rangle \langle p_i \rangle$, i.e., the product of the average colonization rate over all species by the average occupancy of a species, when the number of species goes to infinity. Since this product cannot become negative, this demonstrates that the multi-species competition-colonization version of the Levins cannot be stable by chance alone, and thus a strong

constraint on its parameters is needed for the system to be stable, probably inherited from the constraints on parameters needed for the system to be feasible in the first place. Since the system can also be compared to a competitive Lotka–Volterra multi-species one, this result agrees with other recent theoretical approaches aimed at uncovering conditions for the feasibility of large ecological systems (Bastolla et al. 2005; Dougoud et al. 2018; Stone 2018).

The Levins approach to spatial dynamics assumes a disconnection between local and regional time scales. Interestingly, it allows major disturbances to occur locally, leading to one or several extinctions from a patch. Extensions of the Levins model also have the property of very strong elements along the diagonal of the Jacobian, since all species are subject to a very fundamental constraint: they cannot occupy more than the entire landscape. Overshooting is impossible by construction of the model and the nature of spatial dynamics, and therefore it tends to strongly stabilize dynamics. It does not mean, however, that interactions or the spatial arrangement of patches cannot affect stability. Grilli et al. (2015) has investigated, for instance, the impact of spatial network topology on the stability of metapopulations and found that a random arrangement of patches is more stable and persistent than a regular one. The next step would be to develop a formalism to build Jacobian matrices embedding both the network of interactions and the connectivity matrix. It is unclear for now if May's prediction will hold in a spatialized context with patch dynamics. Since spatial dynamics have often found to be stabilizing, see review in Amarasekare (2008), it may be expected to see a reversal of the stability–diversity relationship. (Gravel et al., 2011a), for instance, found with simulations that more diverse and connected food webs are more persistent than simpler ones, in opposition to May's proposition.

12.7 Conclusion

Metacommunity theory has matured and is ready to undertake a new phase in its development. In conclusion, we propose the following three topics at

the center of a new research program on community assembly in spatially structured environments.

1. Co-distribution: where and when? New methods are proposed to document biotic interactions from co-distribution, e.g., Pollock et al. (2014); Ovaskainen et al. (2017); Harris (2016) but they suffer from inappropriate integration with metacommunity theory (Zurell et al. 2018). Conditions leading to a distinctive signature of interactions on co-distribution need to be established formally.

2. Diversity–stability in metacommunities. Studies are suggesting that the negative diversity–stability relationship proposed by May (1972) may not apply in metacommunities subject to patch dynamics, such as documented in this chapter. Attention should be given to derive Jacobian matrices from patch dynamics and investigate to what extent food web assembly may stabilize dynamics and solve the stability paradox.

3. Empirical investigation of metacommunity models. The model we proposed is simple enough to be parameterized directly from field data, as exemplified by Vissault et al. (in review). Parameterization of the model and test of its quantitative predictions is required to determine if the proposition is an appropriate representation of spatial dynamics and build further on it. In particular, food web models and the assumption of a sequential community assembly need to be explored.

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